
Grokking Algorithms Concepts

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CHAPTER 1

PROBLEMS

1.1 Problem 1:

1.1.1 I can't figure out the binary search implementation!?

What I tried

At first I try to implement binary search without AI help or viewing book's implementation. But this is not worked. After that I use AI to get a correct implementation but my code output was not correct! Finally I used book's implementation in page 9 to correct my different program.

1.2 Problem 2:

1.2.1 I was a little confused about logarithms!

What I tried

1. Read the book's definition for logarithms.
2. Watch some tutorials in KhanAcademy about logarithms.
3. Solve some practices in KhanAcademy about logarithms.

1.3 Problem 3:

1.3.1 I had no idea to understand the concept of permuatation!

What I tried

1. Watch one video about permutation in KhanAcademy
2. Compare this concept to factorial concept ($n!$) and the ambiguity was resolved.

1.4 Problem 4:

1.4.1 I want to inspect call stacks in a Python program that runs a recursive factorial function!

What I tried

1. I used the `strace` tool to inspect call stacks but it is not useful because `strace` just traces system calls in kernel space and does not show me user calls in user space (which Python programs use).
2. I asked AI to get help. It suggests that I use one of these tools:
 - `py-spy`
 - `gdb`
 - `faulthandler`
 - `perf`
3. First I used `py-spy`:

```
py-spy record -o out.svg --pid 66763
```

This is a very long sentence that I want to keep several source lines, but it should render as a single paragraph in the PDF.